

Notes: Liao and Zhang (RFS, 2025)

The Hedging Channel of Exchange Rate Determination

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1 Big picture

- **Question.** How do slowly moving external imbalances get translated into fast and volatile exchange-rate movements?
- **Core idea.** The paper proposes a *currency hedging channel*. Investors and borrowers with net dollar exposures hedge more aggressively when expected FX volatility rises. That hedging demand moves forward prices first, but because intermediaries manufacture forwards by trading spot and money markets, spot rates move too.
- **Setting.** The empirical work uses G-10 currencies, country-level external balance-sheet data, forward and spot exchange-rate data, currency options, a regulation shock from Solvency II, and Federal Reserve swap-line usage during COVID-19.
- **Main theoretical mechanism.** A country with positive dollar debt holdings is effectively “dollar rich.” When volatility rises, its investors want to hedge dollar assets by selling dollars forward and buying home currency forward. Intermediaries meet that demand only at a premium, so the home currency becomes expensive in forward markets and often appreciates in spot markets as well.
- **Main empirical predictions.**
 - Countries with more positive dollar debt imbalances should have more negative cross-currency bases on average.
 - Their currencies’ call options should be more expensive relative to puts.
 - When FX volatility rises, these countries should see their bases become more negative and their spot currencies appreciate.
 - Because volatility mean reverts, these currencies should subsequently depreciate.
- **Main baseline findings.** The data line up closely with the model. Countries with positive dollar debt imbalances have lower unconditional currency excess returns, more negative bases, more expensive domestic-currency calls, stronger appreciation when volatility rises, and predictable reversals later.
- **Key identification message.** The paper does not observe hedge ratios country by country and therefore does not claim a direct causal estimate of hedging demand. Instead, it presents a coherent set of conditional and unconditional moments across several markets that all line up with the same mechanism.
- **Why the paper is useful.** It gives a concrete financial channel that links external balance sheets to exchange-rate behavior, helps explain cross-currency basis variation, term structure effects, option skews, and part of the cross-section of currency premia.
- **Big takeaway.** Exchange rates are not only about macro flows such as trade. They are also about who holds foreign-currency assets and liabilities, how much they hedge, and how costly it is for intermediaries to absorb that hedging demand.

2 Data and measurement (key objects)

2.1 Why the G-10 is the core sample

The empirical analysis focuses on the major freely floating developed-market currencies: Australia (AUD), Canada (CAD), Switzerland (CHF), the euro area (EUR), the United Kingdom (GBP), Japan (JPY), Norway (NOK), New Zealand (NZD), and Sweden (SEK), with the U.S. dollar as the common reference currency.

- These are the most liquid currencies.
- Their forward, option, and money-market data are available at comparable maturities.
- Capital controls and heavy exchange-rate management are limited relative to currencies such as the renminbi or Hong Kong dollar.

Interpretation. The paper wants a setting where forward and option prices are genuinely market-determined and where cross-country comparisons are meaningful.

2.2 External imbalances

The central state variable is a country's **dollar imbalance**, measured from the currency composition of international investment positions. The paper uses the Benetrix et al. data set on cross-border currency exposures and focuses mainly on

$$\text{USD net debt holdings/GDP.} \tag{1}$$

The authors also examine alternative balance-sheet measures:

- dollar NIIP (debt plus equity),
- dollar debt,
- dollar equity,
- and, in some specifications, aggregate net foreign asset positions.

The imbalance data are available quarterly and are forward-filled because country external positions change much more slowly than exchange rates.

Interpretation. The main bet of the paper is that *dollar debt* matters more than broad NFA because fixed-income positions are much more heavily hedged than equity positions.

2.3 Why debt rather than equity

The paper's main mechanism is tied to hedging demand, and that demand is much more common in debt than in equity.

- Institutional investors such as insurers and pension funds often hedge foreign bond exposures heavily because currency risk is large relative to bond returns.
- For equities, exchange-rate risk is smaller relative to total return volatility and is more often left unhedged.

Interpretation. This is why the paper repeatedly finds that dollar *debt* imbalances explain exchange-rate objects much better than dollar *equity* imbalances.

2.4 Cross-currency basis

The main forward-market price measure is the 1-year Libor-based cross-currency basis,

$$b_t^n \approx (f_t^n - s_t^n) + (r_t^D - r_t^n), \quad (2)$$

where f_t^n is the log 12-month forward rate for currency n , s_t^n is the log spot rate, r_t^D is the U.S. Libor rate, and r_t^n is the foreign Libor rate.

A more negative basis means the foreign currency is expensive in forward markets relative to spot after adjusting for interest differentials.

Interpretation. The basis is the cleanest object for the theory because it is literally the price wedge created when intermediaries absorb hedging demand in forward markets.

2.5 Options and volatility variables

The paper uses two option-based objects:

- **ATM implied volatility** at the 1-year horizon as a measure of expected exchange-rate volatility.
- **Risk reversals**, defined as the implied volatility of a 25-delta out-of-the-money call minus the implied volatility of a 25-delta out-of-the-money put.

A positive risk reversal means calls on the domestic currency are expensive relative to puts; a negative risk reversal means puts are expensive.

Interpretation. The options market provides an independent test of the hedging channel. If countries with positive dollar imbalances want protection against home-currency appreciation, their domestic-currency calls should become relatively expensive.

2.6 Spot returns, excess returns, and the term structure

The paper also studies:

- monthly spot exchange-rate returns,
- currency excess returns,
- and the 5-year minus 1-year cross-currency basis spread.

The term-spread object is important because the paper's extended model predicts that longer-dated forwards embed compensation for future states in which hedging demand may spike.

Interpretation. The paper is not content with matching only one market. It tries to show that the same hedging logic organizes forward levels, forward dynamics, option skews, spot movements, term spreads, and swap-line usage.

2.7 Swap-line and regulation case studies

Two case studies broaden the empirical reach:

- **Solvency II**, implemented on January 1, 2016, which imposed capital charges on currency mismatches for European and U.K. insurers and plausibly raised hedging demand.
- **Federal Reserve dollar swap lines** during COVID-19, which can ease the balance-sheet cost of supplying dollar hedges.

Interpretation. These are not the paper’s core cross-sectional panel regressions, but they are useful because they line up with the same mechanism in two very different environments.

2.8 Summary statistics

Table 1 gives a compact sense of magnitudes in the data:

- mean cross-currency basis: -8.39 bps,
- mean absolute basis: 14.02 bps,
- mean currency excess return: 0.99% ,
- mean 5-year minus 1-year basis spread: 2.68 bps,
- mean risk reversal: -0.66 percentage points,
- mean FX volatility: 10.52% ,
- mean USD net debt holdings/GDP: 0.05 ,
- mean USD net equity holdings/GDP: 0.32 .

Interpretation. The average country is only modestly dollar-long in net debt terms, but the cross-country dispersion is large, and that dispersion is what the paper exploits most heavily.

3 Models and empirical specifications (all equations + interpretation)

3.1 Economic mechanism and channel

The model has three moving parts:

- a representative foreign investor with a net dollar debt position,
- an intermediary that manufactures forwards by trading spot and money markets,
- and market-clearing in both forward and spot FX markets.

The key state variables are the country’s external dollar imbalance and expected exchange-rate volatility. The key endogenous object is the hedge ratio.

Interpretation. External imbalances move slowly, but hedge ratios move quickly. That is the bridge between slow macro quantities and fast financial prices.

3.2 Investor problem and optimal hedge ratio

Investor n starts with a net dollar debt position X_n and chooses a hedge ratio h_n . Second-period wealth in domestic currency is

$$W_2^n = h_n X_n R_D F_n + (1 - h_n) X_n R_D S_2^n, \quad (3)$$

where F_n is the forward exchange rate and S_2^n is the future spot rate.

With mean-variance preferences,

$$U_n = E[W_2^n] - \frac{\gamma}{2} \text{Var}(W_2^n), \quad (4)$$

the optimal hedge ratio is

$$h_n = 1 - \frac{E[S_2^n/S_1^n] - (F_n/S_1^n)}{\gamma \text{Var}(S_2^n/S_1^n) X_n S_1^n R_D}. \quad (5)$$

Interpretation. The hedge ratio rises when exchange-rate variance rises, when risk aversion is higher, and when the imbalance is larger. So a volatility shock mechanically creates more hedging demand, especially in countries with large dollar positions.

3.3 Intermediary supply of forwards

The intermediary supplies forwards using spot and funding markets. Its profit from taking position q_n in country n is

$$q_n b_n = q_n \left[(1 + r_D) - \frac{S_n}{F_n} (1 + r_n) \right], \quad (6)$$

where b_n is the cross-currency basis.

The intermediary also faces convex balance-sheet or haircut costs. In reduced form, this gives a basis equation of the form

$$b_n = \kappa G' \left(W - \kappa \sum_m H(-h_m X_m) \right) H'(-h_n X_n). \quad (7)$$

Interpretation. The basis is zero only if hedging demand is zero or intermediation is frictionless. More hedging demand and tighter intermediary constraints both raise the price wedge in forward markets.

3.4 Spot-market clearing and equilibrium

Spot-market clearing is written as

$$\frac{\xi_n}{S_n} - \iota_D - \Theta_D - q_n = 0, \quad (8)$$

where the first three terms capture other trade and financial flows, and q_n is the intermediary's spot-market position used to manufacture forwards.

In equilibrium,

$$q_n = -h_n X_n, \quad (9)$$

so hedging demand in forward markets spills into spot demand through the intermediary's hedging trades.

The paper summarizes equilibrium with four equations:

$$h_n = 1 - \frac{E[S_2^n/S_1^n] - (F_n/S_1^n)}{\gamma \text{Var}(S_2^n/S_1^n) X_n S_1^n R_D}, \quad (10)$$

$$b_n = \kappa G' \left(W - \kappa \sum_m H(-h_m X_m) \right) H'(-h_n X_n), \quad (11)$$

$$S_n = \frac{\xi_n}{\iota_D + \Theta_D - h_n X_n}, \quad (12)$$

$$F_n = \frac{S_n(1 + r_n)}{1 + r_D - \kappa G'(W - \kappa \sum_m H(-h_m X_m)) H'(-h_n X_n)}. \quad (13)$$

Interpretation. Forward demand drives both the basis and the spot rate because the same intermediary balance sheet sits in the middle of both markets.

3.5 Model predictions

The paper emphasizes four predictions.

Proposition 1: Unconditional currency basis.

- Positive external imbalances imply negative bases.
- Negative external imbalances imply positive bases.
- Larger imbalances imply larger absolute bases.

Proposition 2: Conditional currency basis.

- The absolute basis widens when expected exchange-rate volatility rises.
- The widening is stronger in countries with larger imbalances.
- Positive-imbalance currencies become even more overvalued in forward markets when volatility rises; negative-imbalance currencies become more undervalued.

Proposition 3: Spot exchange rates.

- When volatility rises, currencies of positive-imbalance countries appreciate.
- Currencies of negative-imbalance countries depreciate.

Proposition 4: Term structure of basis.

- Longer-maturity basis spreads reflect expected future short-maturity bases.

- The term structure is upward sloping in magnitude when the short basis is narrow and downward sloping in magnitude when the short basis is already wide.

Interpretation. These propositions generate a rich set of testable implications across many assets, not just one headline regression.

3.6 Term-structure extension

In the three-period extension, the two-period basis is a weighted average of current and expected future one-period bases:

$$b_{1,(2)}^n = \frac{b_1^n(1 + r_2^n) + \pi b_{2,L}^n + (1 - \pi)b_{2,H}^n}{2}. \quad (14)$$

Interpretation. If crises and hedging spikes may happen in the future, longer-dated forwards should build in a premium for that risk. That is why the paper links the term structure of cross-currency basis to hedging demand.

3.7 Unconditional empirical specifications

For levels, the paper runs panel regressions of the form

$$y_t^n = \alpha + \beta \text{Imbalance}_t^n + \text{fixed effects} + \varepsilon_t^n, \quad (15)$$

where y_t^n is either the level of the cross-currency basis or the level of the options risk reversal.

These regressions ask whether countries that are more dollar-rich look systematically different in forward and option markets.

Interpretation. This is the unconditional evidence: does the cross section line up with the theory before we even look at volatility shocks?

3.8 Conditional dynamics specification

For changes in bases and options, the paper estimates

$$\Delta y_t^n = \alpha^n + \delta_t + \beta(\text{Imbalance}_t^n \times \Delta \text{FXVol}_t^n) + \Gamma_t^n + \varepsilon_t^n, \quad (16)$$

with currency and date fixed effects and standard errors clustered by currency and date.

Here Δy_t^n is either the change in the basis or the change in the risk reversal.

Interpretation. The cross section of imbalances is slow-moving, so the paper interacts it with *time-varying* volatility to get a dynamic prediction. The volatility shock activates the hedging channel.

3.9 Spot return regression

The same interaction logic is applied to spot returns:

$$\Delta s_t^n = \alpha^n + \delta_t + \beta(\text{Imbalance}_t^n \times \Delta \text{FXVol}_t^n) + \Gamma_t^n + \varepsilon_t^n. \quad (17)$$

The model predicts $\beta < 0$ because higher volatility makes positive-imbalance currencies appreciate relative to negative-imbalance currencies.

Interpretation. This is the central bridge from forward markets to spot markets.

3.10 Forecasting regression

Because volatility mean reverts, the paper predicts that the spot moves generated by hedging demand should partly unwind. It therefore estimates

$$\Delta s_{t,t+\tau}^n = \alpha^n + \gamma(\text{Imbalance}_t^n \times \text{FXVol}_t^n) + \Gamma_t^n + \eta_t^n, \quad (18)$$

for horizons of 3, 6, and 12 months.

The model predicts $\gamma > 0$: if volatility is high today, positive-imbalance currencies should depreciate later as hedges are unwound.

Interpretation. The forecast regression is not just an extra reduced form. It is a sharp dynamic implication of the hedging channel.

3.11 Solvency II differences-in-differences

To study a policy shock to hedging demand, the paper estimates a differences-in-differences regression for the absolute value of the basis:

$$|b^n| = \alpha_0 + \alpha_1 \mathbf{1}_{\text{Solvency II}} + \alpha_2 \mathbf{1}_{\text{Post}} + \alpha_3 (\mathbf{1}_{\text{Solvency II}} \times \mathbf{1}_{\text{Post}}) + |\text{USD Imbalance}_t^n| + \eta_t^n. \quad (19)$$

The treatment group is the euro area and the United Kingdom; the post period begins in January 2016.

Interpretation. If regulation forced more currency hedging, the treated currencies should see larger absolute bases after implementation.

4 Identification and instruments (what makes the estimates credible)

4.1 There is no direct hedge-ratio instrument

The paper is transparent that it does *not* observe country-level hedge ratios and therefore does not offer a single clean causal estimate of hedging demand. The strategy is instead one of *triangulation*:

- forward levels,
- option levels,
- forward dynamics,
- spot dynamics,
- return forecasting,

- term-structure patterns,
- Solvency II,
- and swap-line use

all line up with the same external-imbalance-by-volatility logic.

Interpretation. The identification here is closer to a structural moment-matching argument than to a single quasi-experiment.

4.2 What identifies the main interaction estimates

The core empirical variation comes from two ingredients:

- persistent cross-country differences in dollar debt holdings,
- time-series variation in expected FX volatility.

The key coefficient asks whether volatility shocks have systematically different effects in countries with different balance-sheet positions.

Because the regressions include currency fixed effects and time fixed effects, identification is effectively from *relative* responses across currencies within the same month.

Interpretation. The paper is not asking whether volatility matters on average. It is asking whether volatility matters *differently* depending on who is dollar-long and who is dollar-short.

4.3 Why reverse causality is limited

Could exchange-rate movements themselves mechanically change the imbalance measure and generate spurious results? The paper's answer is largely no, for two reasons:

- the main imbalance measures are slow-moving stock variables that are updated only annually or quarterly and then forward-filled;
- the dynamic regressions use volatility changes and fixed effects, not contemporaneous portfolio-choice microdata.

Interpretation. The identifying assumption is that short-run changes in implied volatility do not endogenously create large changes in underlying national dollar debt positions at monthly frequency.

4.4 Why debt beats broad NFA

A major credibility check is that the results are stronger for dollar debt than for total net foreign assets.

- In the basis and option *levels* regressions, dollar debt is strongly significant while dollar equity is not.
- In the spot-return regressions, interacting total NFA with volatility produces much weaker and statistically insignificant coefficients once dollar debt is included.

Interpretation. This is exactly what the hedging channel predicts. If the mechanism were simply “rich creditor countries behave differently,” broad NFA should dominate. Instead, the hedged part of the external balance sheet matters most.

4.5 Options and term structure as additional validation

The options and term-structure results are especially useful because they are hard to rationalize with a narrow spot-only story.

- The same countries that have more negative bases also have more expensive domestic-currency calls relative to puts.
- Countries with negative imbalances have upward-sloping basis term structures, while countries with positive imbalances have downward-sloping ones.

Interpretation. Matching these extra moments makes the hedging explanation much more persuasive than a one-market reduced form would be.

4.6 Solvency II as a quasi-shock to hedging demand

Solvency II gives the paper a natural demand-side event: insurers in the euro area and the U.K. were penalized for currency mismatches and therefore had reason to hedge more after 2016.

This is useful because the treated currencies’ basis magnitudes rise relative to the rest of the G-10 around the reform date.

The paper is also careful about the limits:

- implementation was anticipated, so there are pre-trends;
- European interest differentials also changed around the same time.

Interpretation. Solvency II is suggestive rather than definitive, but it points in exactly the direction the hedging channel would predict.

4.7 Swap-line case study and mechanism discrimination

The swap-line evidence is powerful because the paper’s mechanism makes a counterintuitive prediction: countries with *positive* dollar debt holdings may use dollar swap lines more, even though debtor countries are often thought to need dollars most urgently.

The idea is that dollar-rich countries need dollars to help intermediaries produce local-currency forwards for hedgers. Dollar-poor countries instead want to buy dollars forward, so direct dollar provision is less helpful for their hedging need.

Interpretation. This is a clean mechanism distinction between funding demand and hedging demand.

4.8 What the paper can and cannot distinguish

The paper can show that a hedging-based mechanism organizes a broad set of exchange-rate facts. It cannot fully separate:

- investor hedging demand from all other balance-sheet-demand channels,
- direct hedging demand from confidence effects of swap-line announcements,
- or country-level hedge-ratio movements from other country-specific institutional features.

Interpretation. The paper's strongest claim is that the hedging channel is quantitatively important and empirically relevant, not that it is the only force in exchange-rate determination.

5 Tables and figures

5.1 Figure 1: Japanese life-insurer hedge ratio

Read:

- The figure plots a hedge ratio for large Japanese life insurers against the currency volatility index.
- The hedge ratio is high on average and rises when exchange-rate volatility rises.

Economic message. This is the paper's motivating micro fact: hedging demand is time varying and countercyclical. That is what allows a slow-moving external imbalance to generate fast-moving exchange-rate effects.

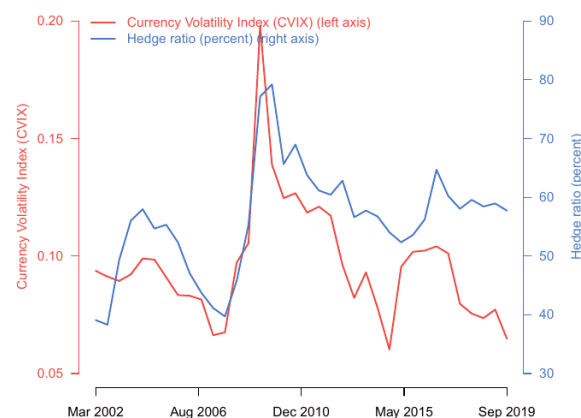


Figure 1

5.2 Table 1 and Figure 2: unconditional basis and option evidence

Read:

- Table 1 shows that the average basis is negative and the average absolute basis is economically meaningful at 14 bps.

- Figure 2, Panel A shows a strong negative cross-sectional relation between average cross-currency bases and average dollar debt holdings.
- Figure 2, Panel B shows a strong positive relation between risk reversals and dollar debt holdings.
- Table 2 reports the same message in regressions: the coefficient on dollar debt is about -25.6 in the basis regressions, while the equity coefficients are weak.
- Table 3 reports that a 10 percentage point higher dollar debt imbalance is associated with roughly a 0.275 percentage point higher call-minus-put pricing wedge.

Economic message. Before looking at volatility shocks, the cross section already looks like the model. Dollar-rich countries pay more to hedge appreciation risk, both in forwards and in options.

Table 1
Summary statistics

	Mean	SD	Min	Max
Cross-currency basis (bps)	-8.39	17.93	-92.15	42.11
Abs. cross-currency basis (bps)	14.02	13.98	0.01	92.15
Currency excess returns (pct)	0.99	10.67	-37.25	35.1
5-yr minus 1-yr basis spread (bps)	2.68	11.28	-48.95	60.75
Risk-reversal (pct)	-0.66	1.55	-7.28	9.85
FX volatility (pct)	10.52	2.84	4.78	22.98
Δ FX volatility (pct)	0	0.06	-0.18	0.56
USD NFA / GDP	0.37	0.38	-0.25	1.71
USD net debt holdings / GDP	0.05	0.27	-0.32	0.99
USD net equity holdings / GDP	0.32	0.21	0.04	0.99

The sample comprises monthly data for all G-10 currencies (excluding the USD) between January 2000 and December 2021. A currencies' cross-currency bases is the spread between the exchange rate implied currency risk-free rate and the actual risk-free rate. The annualized currency excess return is the difference between the log 12-month forward rate and the log spot exchange rate in 12 months. NIIP, Debt, FDI, Equity, and GDP are measured quarterly and provided by the International Financial Statistics (IFS) from the IMF.

behavior of hedge ratios for a large sample of countries. Therefore, instead of

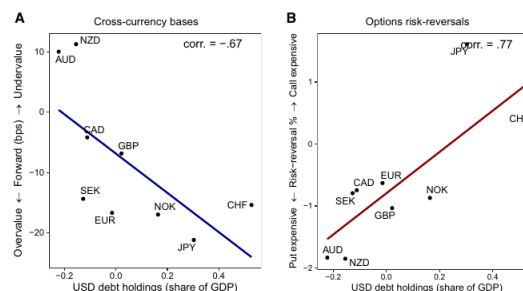


Figure 2
Dollar imbalances, cross-currency bases, and options risk-reversals

This figure presents unconditional relationships between average cross-currency bases, currency risk reversals and dollar imbalances. Panel A shows the relationship between average cross-currency bases and average dollar imbalances. The slope of the regression line is -32.77 (s.e. = 13.68). Panel B shows the relationship between average risk reversals and average dollar imbalances. A risk reversal captures the relative pricing of calls and puts on currencies as measured by the difference in implied volatility between the 25-delta call minus put. The slope of the regression line is 3.34 (s.e. = 1.04). Our sample ranges from January 2000 to December 2021.

Table 2
External imbalances and cross-currency bases

	Cross-currency basis (basis points)					
	(1)	(2)	(3)	(4)	(5)	(6)
USD NIIP	-17.50** (7.76)	-17.43* (9.70)				
USD debt			-25.56** (11.70)	-24.43* (12.57)		
USD equity					-17.64 (12.08)	-14.48 (21.70)
Fixed effects		Month		Month		Month
Num. obs.	2,363	2,363	2,363	2,363	2,363	2,363
R-sq. (full)	.14	.37	.14	.38	.04	.28
R-sq. (proj)	.14	.15	.14	.17	.04	.03

This table presents panel regressions of monthly cross-currency bases on various measures of external imbalances. USD NIIP captures a region's dollar denominated debt plus equity imbalance. USD debt captures the region's dollar denominated debt imbalance, and USD equity captures a region's dollar denominated equity imbalance. All measures are normalized by GDP. The external imbalance measures are updated at an annual frequency. The sample period is from 2000 to 2021. Standard errors are clustered by currency. * $p < .1$; ** $p < .05$; *** $p < .01$.

5.3 Table 4 and Figure 3: conditional dynamics of forwards and options

Read:

- Table 4 interacts dollar imbalances with changes in FX volatility.
- In the basis regressions, the interaction coefficients are negative and significant: when volatility rises, positive-imbalance countries' bases become more negative.

- In the options regressions, the interaction coefficients are positive and significant: when volatility rises, domestic-currency calls become relatively more expensive in positive-imbalance countries.
- Figure 3 shows that option skews widen sharply in crisis episodes and do so in directions that differ across currencies depending on their balance-sheet positions.

Economic message. This is the heart of the paper's dynamic evidence. Volatility shocks do not move all currencies the same way; they interact with the external balance sheet exactly as the hedging channel predicts.

Table 4
Cross-currency bases and risk-reversal dynamics

	Currency basis (bps)		Risk reversals (pps)	
	(1)	(2)	(3)	(4)
USD imbalance $\times$ Δ FX vol.	-1.74** (0.56)		0.38* (0.17)	
USD imbalance $\times$ Δ Global FX vol.		-1.67** (0.67)		0.40** (0.17)
USD imbalance	0.73 (0.63)	0.72 (0.66)	0.00 (0.02)	0.01 (0.04)
Δ FX vol.	-0.31 (0.25)		-0.12*** (0.03)	
Obs.	2,362	2,362	2,363	2,363
R-sq. (full model)	.46	.46	.45	.45
R-sq. (proj model)	.03	.02	.14	.15

The following table presents panel regressions of monthly changes in cross-currency bases and risk-reversals on dollar debt imbalances and measures of exchange rate volatility. Cross-currency bases are measures in basis points and risk-reversals are measured in percentage points. Δ FX vol. captures changes in country specific FX volatility, and Δ Global FX vol. captures changes in global FX volatility defined as the average change in FX volatility across all countries. All regressions include month and currency fixed effects. Standard errors are clustered by currency and date. *p<.1; **p<.05; ***p<.01.

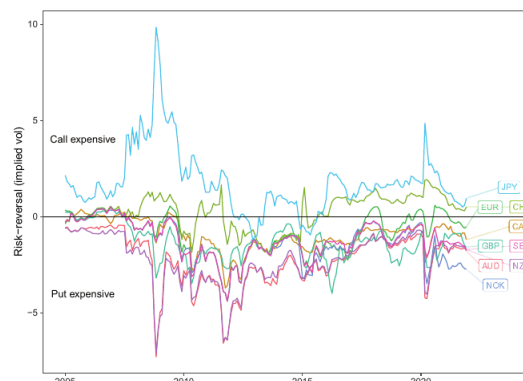


Figure 3
Currency options risk-reversals
This figure presents the time series of relative pricing of calls and puts on G-10 currencies as measured by their risk-reversals. The sample runs from January 2005 to December 2021.

of cross-currency bases.²⁸ First, options risk reversals increased in magnitude

5.4 Table 5 and Table 6: spot exchange rates and forecasting

Table 5
Spot exchange rate dynamics

	(1)	(2)	(3)	(4)
USD imbalance $\times$ Δ FX vol.	-1.15* (0.55)		-1.44* (0.64)	
USD imbalance $\times$ Δ Global FX vol.		-1.16* (0.56)		-1.43* (0.67)
NFA $\times$ Δ FX vol.			0.16 (0.31)	
NFA $\times$ Δ Global FX vol.				0.15 (0.29)
USD imbalance	-0.08 (0.26)	-0.10 (0.27)	-0.08 (0.29)	-0.09 (0.29)
Δ FX vol.	0.64*** (0.17)		0.62*** (0.18)	
NFA			0.00 (0.08)	0.01 (0.09)
Obs.	2,363	2,363	2,354	2,354
R-sq. (full model)	.62	.61	.61	.61
R-sq. (proj model)	.07	.04	.07	.05

This table presents panel regressions of monthly changes in spot exchange rate returns on dollar debt imbalances and measures of exchange rate volatility. See the notes of Table 4 for variable definitions. All regressions include month and currency fixed effects. Standard errors are clustered by currency and date. *p<.1; **p<.05; ***p<.01.

more expensive for countries with positive imbalances, and put options become

Table 6
Exchange rate forecasting regressions

	3 month	6 month	12 month
USD imbalance $\times$ FX vol	0.93 (0.67)	2.33* (1.20)	4.00* (2.23)
FX vol	-0.49** (0.20)	-1.07*** (0.36)	-1.62*** (0.58)
USD imbalance	1.34 (1.68)	3.07 (3.06)	5.83 (5.22)
Obs.	2,336	2,309	2,255
R-sq.	.01	.02	.03

This table presents the results of exchange rate return forecasting regressions. At time horizons of 3, 6, and 12 months, we regress exchange rate returns on FX vol, USD imbalance, and the interaction term USD imbalance $\times$ FX vol at the start of the period. We include a currency fixed effect in all specifications. We compute Newey-West standard errors with lags equal to 1.5 times the return horizon. *p<.1; **p<.05; ***p<.01.

where $s_{t,t+\tau}^n$ is the log spot exchange rate return between period t and $t+\tau$, α^n continues to capture a currency fixed effect, Ξ^n is a vector of controls. Notably, we interact the country n imbalance with the level of the currency n volatility, rather than the change in volatility, because a higher than average exchange rate volatility level will likely revert in the future. γ is the coefficient of interest, which should be positive.

Table 6 shows the results of exchange rate forecasting regressions. For forecast horizons

Read:

- Table 5 shows that currencies of positive dollar-debt countries appreciate when expected FX volatility rises. The magnitude is about 1.15 percentage points for a country with dollar debt holdings equal to GDP after a one-standard-deviation volatility increase.

- Once the paper controls for interactions involving aggregate NFA, the dollar-debt interaction remains significant and the broad NFA interaction becomes weak.
- Table 6 shows the reversal side of the mechanism: high volatility today predicts later depreciation for positive-imbalance currencies, about 2.33% over six months and 4.00% over 12 months.

Economic message. The spot-market results are what transform the paper from a basis paper into a full exchange-rate paper. Hedging pressure moves spot rates now and helps forecast reversals later.

5.5 Figure 4, Table 7, and Figure 5: currency premia and term structure

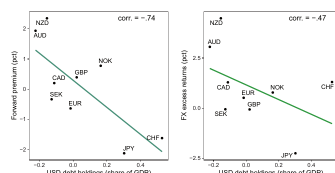


Figure 4
Dollar imbalances, forward premia, and currency excess returns
The left-hand-side panel of this figure plots the average 12-month forward premia of each G-10 currency against the countries' average dollar imbalances. The slope of the regression line is -4.52 (i.e. $\approx -1.5\%$). The right-hand-side panel plots the average 12-month currency excess returns of each G-10 currency against countries' dollar debt imbalances. The slope of the regression line is -3.70 (i.e. $\approx -2.6\%$).

4.5 Discussion: Implications for currency risk premiums

	Level (bps)		Changes (bps)	
	(1)	(2)	(3)	(4)
USD imbalance	-19.55***	-20.60**	0.00	0.00
	(7.32)	(7.30)	(0.12)	(0.12)
USD imbalance $\times \Delta$ FX Vol			0.34*	0.34
			(0.16)	(0.16)
USD imbalance $\times \Delta$ Global FX vol			0.61	0.35
			(0.12)	(0.12)
Δ FX vol			0.12	0.12
			(0.18)	(0.18)
Date F.E.		Y	Y	Y
Currency F.E.		Y	Y	Y
Num. obs.	2,360	2,360	2,347	2,347
R-sq. (full model)	.21	.37	.34	.34
R-sq. (grey model)	.21	.26	.03	.00

This table presents panel regressions of levels and changes in the 5-year minus 1-year currency basis spread on dollar debt imbalances and measures of exchange rate volatility. Columns 1 and 2 regress the level of the basis spread on dependent variables, while columns 3 and 4 regress changes in the basis spread. See Table 6 for additional variable definitions. Standard errors are clustered by currency and date. * $p < .1$; ** $p < .05$; *** $p < .01$.

have been highlighted previously by Della Corte, Riddiough, and Sarno (2016), who showed cross-sectional variation in currency excess returns is related

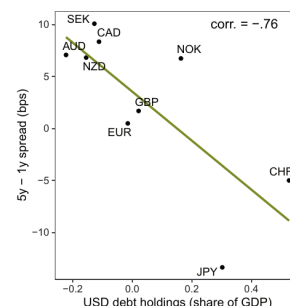


Figure 5
Term structure of cross-currency basis
This figure plots the average spread between the 5-year and the 1-year cross-currency basis against a dollar imbalances for the G-10 currencies from January 2000 to December 2021.

Read:

- Figure 4 shows that countries with more positive dollar debt holdings have lower forward premia and lower currency excess returns on average.
- Table 7 shows that the 5-year minus 1-year basis spread is more negative for countries with larger positive imbalances, and that the term structure moves systematically with volatility.
- Figure 5 visualizes the negative relation between dollar debt holdings and the 5-year minus 1-year basis spread.

Economic message. The same balance-sheet mechanism that explains basis levels and spot dynamics also helps explain why some currencies look safe and why term premia differ across currencies.

5.6 Figure 6 and Table 8: Solvency II

Read:

- Figure 6 shows the euro and the pound experienced a sharp increase in absolute basis magnitude around the implementation of Solvency II relative to other G-10 currencies after controlling for imbalances.

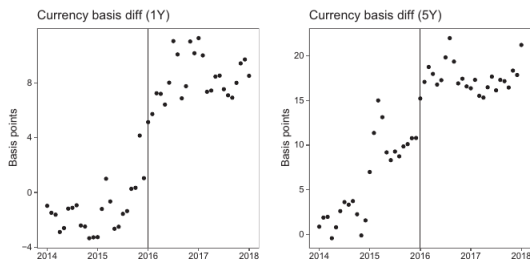


Figure 6
Solvency II and cross-currency bases
These figures present the average difference between the magnitude of cross-currency bases in the United Kingdom and the euro area against the other G-10 currencies after controlling for differences in dollar imbalances. We compute the residuals from a regression of the magnitude cross-currency bases on the magnitude of dollar imbalances. We then compute the difference between the residuals for the euro area and United Kingdom against the other G-10 currencies. The left-hand-side figure shows results for the 1-year tenor, and the right-hand-side figure shows results for the 5-year tenor.

Thus, the implementation of Solvency II presents an opportunity to study the impact of an increase in hedging demand on the currency market. This increase

Table 8
Solvency II and cross-currency basis

	Monthly data		Collapsed data	
	1 year	5 year	1 year	5 year
Solvency 2 × Post	8.96** (3.47)	10.03** (4.72)	8.87*** (2.22)	10.15** (3.67)
Solvency 2	5.54 (3.56)	1.63 (2.11)	5.67*** (0.85)	1.46 (2.65)
Post	-3.17 (6.67)	1.39 (9.38)	-3.47 (5.48)	1.80 (8.99)
USD imbalance	31.10*** (10.05)	60.12*** (23.18)	29.52*** (7.13)	62.22* (28.89)
Constant	11.67*** (2.57)	7.68 (5.73)	12.06*** (2.18)	7.16 (6.86)
Num. obs.	441	441	18	18
R-sq. (full model)	.28	.27	.35	.30
R-sq. (proj model)	.28	.27	.35	.30

This table presents results of a differences-in-differences regression around the implementation of Solvency II. Specifically, we run the following regression:

$$|b^R_t| = \alpha_0 + \alpha_1 \mathbb{I}_{\text{Solvency 2}} + \alpha_2 \mathbb{I}_{\text{Post}} + \alpha_3 (\mathbb{I}_{\text{Solvency 2}} \times \mathbb{I}_{\text{Post}}) + |\text{USD Imbalance}_t|^{\eta} + \eta_t^R$$

where $|\cdot|$ represents the absolute value function, $\mathbb{I}_{\text{Solvency 2}}$ is an indicator for whether the currency region was affected by the Solvency II regulation and $\mathbb{I}_{\text{Post}}$ is an indicator equal to 1 for months including and after January 2016. We restrict the sample to observations in a two-year window around January 2016 (January 2014 to December 2017). In the first two columns, we run a regression above using data at the monthly frequency. In the last two columns, we run the same specification, but we collapse the data into a pre- and post-implementation period. Standard errors are clustered by currency and date. *p < .1; **p < .05; ***p < .01.

- Table 8 estimates the post-treatment effect at about 9 bps for the 1-year tenor and 10 bps for the 5-year tenor in the monthly differences-in-differences specification.
- The 5-year figure rises earlier than the 1-year figure, consistent with anticipation and with longer maturities pricing future hedging demand sooner.

Economic message. A regulation that made currency mismatch more expensive appears to have raised hedging demand and widened bases where the theory says it should.

5.7 Figure 7: swap-line usage during COVID-19

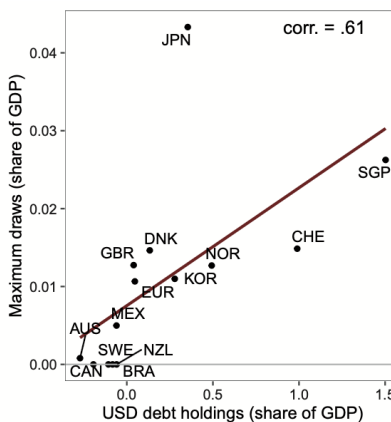


Figure 7
Dollar imbalances and swap lines during COVID-19
This figure plots the maximum swap line draw by each central bank between March 2020 and July 2020 against the country's dollar imbalance. The regression line has a slope of 0.02 (s.e. = 0.01).

Read:

- Figure 7 shows that maximum Fed swap-line draws during March–July 2020 are larger in countries with more positive dollar debt holdings.
- Japan is the most striking case, with both a very positive imbalance and very large draw.

- The paper also notes that long-maturity 84-day operations dominated usage during COVID-19, unlike in the GFC when short-maturity operations were much more important.

Economic message. This is the paper’s most distinctive case-study result. Swap lines seem to have supported hedging demand, not just classic bank funding demand.

6 Short summary

- The paper proposes that time-varying currency hedging links slowly moving external imbalances to fast-moving exchange rates.
- Positive dollar debt imbalances create demand to sell dollars forward and buy home currency forward when volatility rises.
- Limited intermediary balance sheets turn that demand into negative cross-currency bases, option skews, and spot appreciation.
- The unconditional cross section of bases, risk reversals, forward premia, and currency excess returns all lines up with dollar debt holdings.
- The conditional dynamics also line up: volatility shocks move bases, options, spot returns, and the basis term structure differently depending on the sign and size of the imbalance.
- Solvency II provides suggestive event evidence that higher required hedging widened euro and pound bases.
- Fed swap-line usage during COVID-19 is strongest in dollar-rich countries, which fits a hedging-demand interpretation.
- The paper does not directly observe hedge ratios by country, so its case is cumulative rather than based on a single causal design.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes that a country’s external dollar debt position is strongly related to the way its currency behaves in forward, option, and spot markets, especially when exchange-rate volatility moves. The common mechanism is time-varying currency hedging interacting with intermediary frictions.

7.2 Economic intuition

The core intuition is simple: a country that owns lots of dollar bonds is exposed to home-currency appreciation. When volatility rises, investors in that country want more insurance. They buy that insurance in forward and option markets, and intermediaries must use balance sheets and spot trades to provide it. Prices move because intermediation is costly.

7.3 Takeaway

This paper is best read as a modern asset-pricing and international-finance bridge. It shows that exchange-rate behavior can be understood through the same balance-sheet logic that has been so useful in other markets: who wants hedges, who supplies them, and how constrained the suppliers are. That makes the exchange rate less of a macro residual and more of a market-clearing asset price shaped by financial structure.